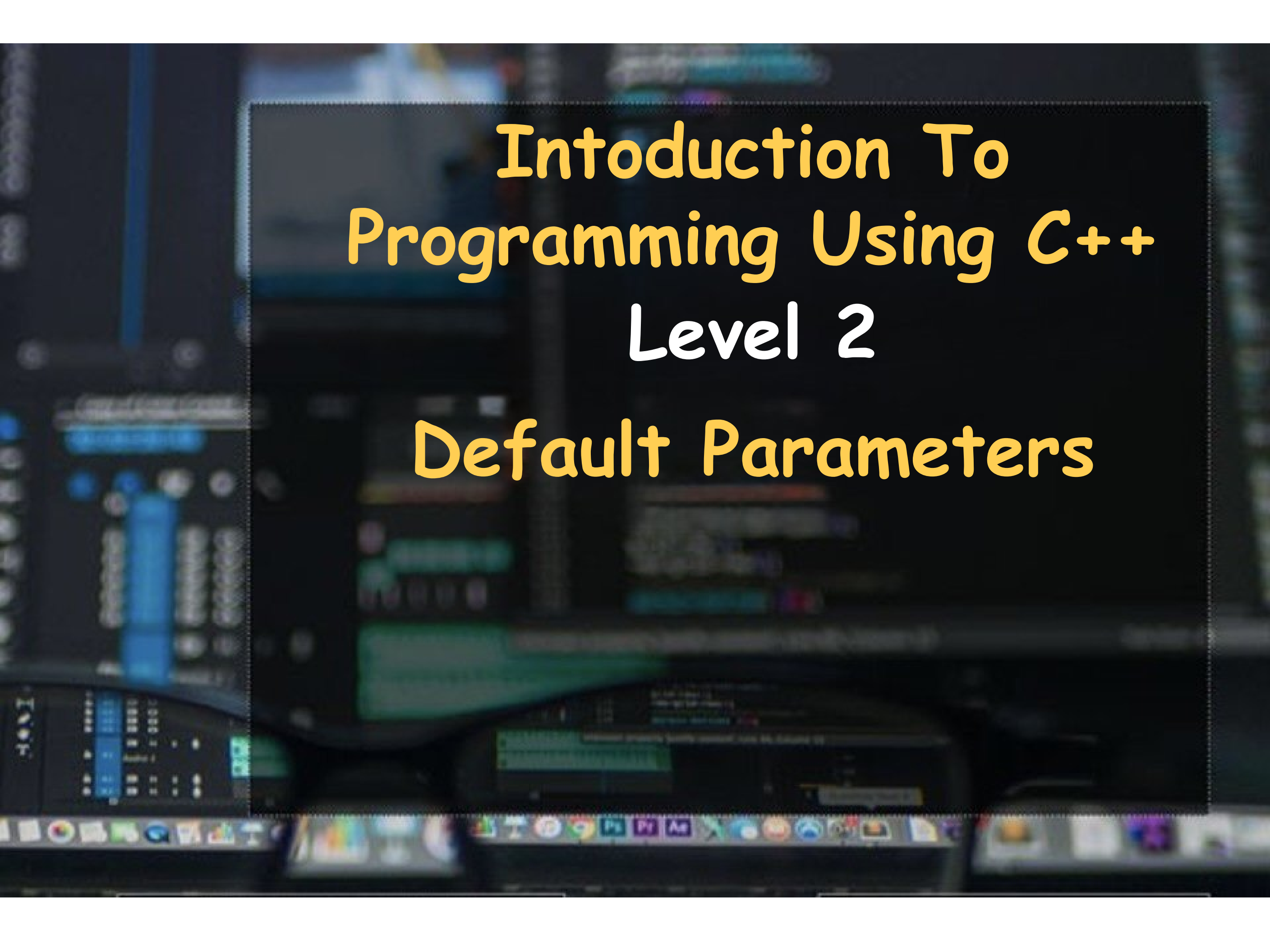
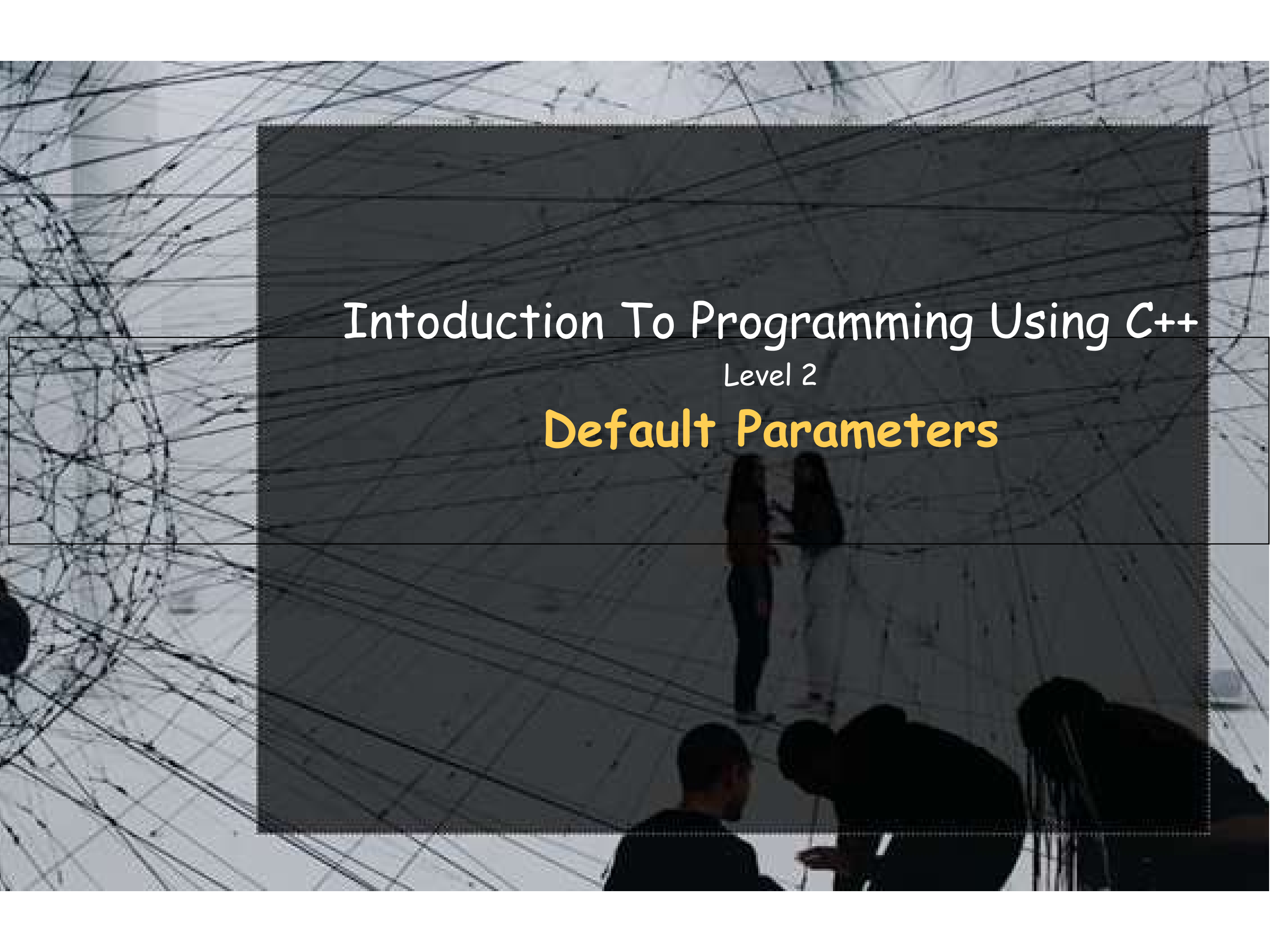




Introduction To Programming Using C++ Level 2

Default Parameters



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Level 2

Default Parameters

Definition

- A Default Parameter in C++ is a parameter that has a predefined value.
- It is an optional parameter. If the user does not provide a value for it when calling the Function, the predefined Default Value is used.
- In programming, Default means the value that is automatically used when another value is not provided.

Explanation

■ In C++, if a Function has Default Parameters, the user does not have to provide values for all Parameters.

■ If a value is not provided, C++ uses the Default Value.

■ If a value is provided, the provided value replaces the Default Value.

■ In the following Function:

```
int MySum(int a, int b, int c = 0, int d = 0)
{
    return (a + b + c + d);
}
```

■ `a` and `b` are required Parameters.

■ `c` and `d` are optional Parameters.

■ The Default Value of both `c` and `d` is `0`.

Procedures

1. Define the required Parameters first.
2. Add Default Values to the optional Parameters.
3. When calling the Function, provide only the values you need.
4. If a Default Parameter is not given a value, C++ uses its Default Value.

💡 Example

```
int MySum(int a, int b, int c = 0, int d = 0)
{
    return (a + b + c + d);
}
int main()
{
    cout << MySum(10, 20) << endl;
    cout << MySum(10, 20, 30) << endl;
    cout << MySum(10, 20, 30, 40) << endl;

    return 0;
}
```

➡ Result

30
60
100

Note

◆ All Parameters with Default Values must be placed at the end of the Parameter List.

◆ This is invalid:

```
int Test(int a = 1, int b);
```

✗ `b` is a required Parameter after a Default Parameter.

◆ This is valid:

```
int Test(int a, int b = 1);
```

✓ The required Parameter comes first, followed by the Default Parameter.

⊘ Warning

◆ Once you start using Default Parameters, the Parameters to their right must also have Default Values.

◆ For example, this is invalid:

```
int Test(int a = 1, int b = 2, int c);
```

◆ `c` is a required Parameter after Default Parameters.



Comparison

▨ With Default Parameters

- ✓ Some Parameters can be optional.
- ✓ Missing values are automatically replaced with Default Values.
- ✓ Can reduce the need for multiple Function Overloads.

▤ Without Default Parameters

- ✓ The caller must provide all required Parameters.
- ✓ Different combinations may require Function Overloading.
- ✓ Can result in more Function Definitions for similar cases.

✦ Summary

- ✓ A Default Parameter has a predefined value.
- ✓ If no value is provided, the Default Value is used.
- ✓ If a value is provided, it replaces the Default Value.
- ✓ Required Parameters must come before Default Parameters.
- ✓ Default Parameters can reduce the need for Function Overloading.

Thank you 😊